

Natural Language Processing (NLP) Lab

Module 2: Text Acquisition and Preprocessing Pipeline

Natural Language Processing is a subfield of artificial intelligence.

It enables computers to understand, interpret, and manipulate human language.

Key stages in text processing:

1. Text acquisition from various file formats such as PDF, DOCX, and HTML.
2. Text extraction and character encoding normalization (UTF-8, NFC).
3. Noise removal, regular expression filtering, and tokenization.

Text extraction from PDF files can encounter challenges with layouts, multi-column formatting, headers, footers, and font encodings.

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Page 2: Downstream NLP Tasks & Applications

Once clean text is extracted, it can be used in machine learning models. Common downstream tasks include text classification, information retrieval, named entity recognition (NER), and large language model fine-tuning.

Effective text cleaning ensures high model quality and reduced noise.

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